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Electronic module, imaging unit and equipment

Abstract

A printed wiring board has a first conductive layer including a second electrode group bonded to a first electrode group of a flexible printed circuit board, and a second conductive layer, and a plane including the second conductive layer includes a first region including a reference region set in a bonded portion, a second region via a first boundary, and a third region via a second boundary, and a first boundary portion is positioned within a range of 0.5 times or more and 5 times or less a length of a short side of four sides in a prescribed direction, and a second boundary portion is positioned within a range of 0.5 times or more and 5 times or less the length of the short in a prescribed direction, and a density of the second conductive layer in the first region is higher than that in the second region.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9421628	12/2015	Higuchi	N/A	N/A
10582615	12/2019	Noguchi	N/A	N/A
10897820	12/2020	Noguchi	N/A	N/A
11595557	12/2022	Okada	N/A	N/A
11765832	12/2022	Hasegawa	N/A	N/A
11877392	12/2023	Hasegawa	N/A	N/A
11895793	12/2023	Noguchi	N/A	N/A
2020/0154016	12/2019	Takakura	N/A	H05K 1/0218
2020/0170112	12/2019	Matsumaru	N/A	N/A
2021/0014391	12/2020	Uemura et al.	N/A	N/A
2022/0020807	12/2021	Hasegawa	N/A	N/A
2022/0346219	12/2021	Hasegawa et al.	N/A	N/A
2022/0353995	12/2021	Tsuboi	N/A	H05K 3/3452
2023/0188822	12/2022	Okada	N/A	N/A
2023/0223713	12/2022	Hasegawa	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2008-140916	12/2007	JP	N/A
2011-171352	12/2010	JP	N/A
2020-120106	12/2019	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an electronic module, an imaging unit, and equipment.

Description of the Related Art

(2) A technique of connecting a flexible printed board to a rigid printed wiring board by soldering is known. In such technical field, a heat generation problem that heat generation due to a resistance loss occurs at a solder bonded portion by improving performance of equipment and speeding up a signal flowing through a flexible printed board becomes significant.

(3) Further, influence of heat on an active component such as ICs (Integrated Circuits), crystals, and the like arranged around the flexible printed board has been considered as a problem.

(4) With the narrowing of the pitch of the flexible printed board, it is difficult to radiate heat by increasing the copper foil area of the solder bonded portion, and it is required to radiate heat not only in the surface layer but also in the inner layer of the substrate. At the same time, suppression of influence of heat on a peripheral component is also an important problem.

(5) Japanese Patent Application Laid-Open No. 2020-120106 discloses an imaging unit in which a flexible printed circuit board is connected to a printed wiring board via solder.

(6) In the technique described in Japanese Patent Application Laid-Open No. 2020-120106, there is room for improvement in the heat dissipation structure of the printed wiring board and the flexible printed circuit board.

SUMMARY OF THE INVENTION

(7) Accordingly, it is an object of the present invention to provide a technique capable of reducing the influence of heat on an electronic component mounted on a printed wiring board and improving heat dissipation in an electronic module in which a flexible printed board is connected to a printed wiring board by soldering.

(8) According to one aspect of the present invention, there is provided an electronic module including: a flexible printed circuit board having a first electrode group; a printed wiring board having a second electrode group; and an electronic component mounted on the printed wiring board, wherein the printed wiring board includes a first conductive layer formed to include the second electrode group and a second conductive layer formed under the first conductive layer, wherein the first electrode group and the second electrode group are bonded by solder, wherein a plane including the second conductive layer includes a first region, a second region in contact with the first region via a first boundary, and a third region in contact with the second region via a second boundary in a plan view with respect to the plane, wherein the first region includes a reference region having a contour of a rectangle, and the reference region is set such that four sides of the rectangle are circumscribed to a bonded portion where the first electrode group and the second electrode group are bonded to each other in the plan view, wherein a first boundary portion, which is at least a part of the first boundary, is positioned within a range of 0.5 times or more and 5 times or less a length of a short side of the four sides in one direction from one side of the four sides of the reference region toward the outside of the reference region, wherein a second boundary portion, which is at least a part of the second boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in the one direction from the first boundary portion, and wherein a density of the second conductive layer in the first region is higher than a density of the second conductive layer in the second region, and the electronic component is arranged on the third region.

(9) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A, FIG. **1**B, and FIG. **1**C are a schematic diagram illustrating an electronic module according to a first embodiment.
(2) FIG. **2** is a schematic diagram illustrating an electronic module according to a second embodiment.
(3) FIG. **3** is a schematic diagram illustrating an electronic module according to a third embodiment.
(4) FIG. **4**A, FIG. **4**B, and FIG. **4**C are a schematic diagram illustrating an electronic module according to a fourth embodiment.
(5) FIG. **5**A and FIG. **5**B are a schematic diagram illustrating an imaging unit according to a fifth embodiment.
(6) FIG. **6** is a schematic diagram illustrating electronic equipment according to a sixth embodiment.
DESCRIPTION OF THE EMBODIMENTS
(7) Embodiments for implementing the present invention will be described in detail below with reference to the drawings. Note that the present invention is not limited to the embodiments described below and can be changed as appropriate within the scope not departing from the spirit of the present invention. Further, in the drawings described below, components having the same function are labeled with the same reference symbols, and the description thereof may be omitted or simplified.
First Embodiment
(8) An electronic module **100** according to a first embodiment will be described with reference to FIGS. **1**A to **1**C. FIG. **1**A is a top plan view of the electronic module **100**. FIG. **1**B is a plan view illustrating a second conductive layer **13** of a second layer from a surface layer when the electronic module **100** is viewed from the top. FIG. **1**C is a cross-sectional view taken along a line A-A′ in FIG. **1**A.
(9) As illustrated in FIGS. **1**A to **1**C, the electronic module **100** according to the present embodiment includes a printed wiring board **10**, a flexible printed circuit board **1**, and peripheral components **40**. As will be described later, the peripheral components **40** and the flexible printed circuit board **1** are connected to the printed wiring board **10** by solder bonding via the solder **5**, respectively.
(10) Here, the coordinate axes and directions of the X-axis, the Y-axis, and the Z-axis of an XYZ coordinate system, which is an orthogonal coordinate system used in the following description, are defined. First, an axis perpendicular to the main surface of the printed wiring board **10** is defined as the Z axis. Further, an axis parallel to the main surface of the printed wiring board **10** and along the long sides of a pair of parallel edges of the printed wiring board **10** is defined as the X axis. An axis orthogonal to the X-axis and the Z-axis is defined as the Y-axis. In the XYZ coordinate system in which the coordinate axes are defined in this manner, a direction along the X axis is defined as an X direction, and among the X directions, a direction from one end of the printed wiring board **10** to the other end is defined as a +X direction, and a direction opposite to the +X direction is defined as a −X direction. Further, a direction along the Y axis is defined as a Y direction, and among the Y directions, a direction from the right side to the left side with respect to the +X direction is defined as a +Y direction, and a direction opposite to the +Y direction is defined as a −Y direction. Further, a direction along the Z axis is defined as a Z direction, and among the Z directions, a direction to

the side of surface to which the flexible printed circuit board **1** of the printed wiring board **10** is connected from the opposite side is defined as a +Z direction, and a direction opposite to the +Z direction is defined as a −Z direction. FIGS. **1**A and **1**B are plan views of the electronic module **100** as viewed in the −Z direction from the side where the flexible printed circuit board **1** is connected, and FIG. **1**C is a cross-sectional view of the electronic module **100** as viewed in the −Y direction.
(11) The flexible printed circuit board **1** includes a flexible base **2**, a flexible wiring layer **3**, and a coverlay **4**. The flexible printed circuit board **1** includes one or more conductive layers as the flexible wiring layer **3**. In the flexible printed circuit board **1**, the flexible wiring layer **3** as a conductive layer can be stacked on the flexible base **2** as an insulating layer via the flexible base **2**. The flexible wiring layer **3** of a surface is covered with a coverlay **4**. In the present embodiment, a case where the flexible wiring layer **3** in the flexible printed circuit board **1** is a single layer will be described, but the flexible wiring layer **3** is not limited to a single layer and may be multiple layers.
(12) Although the flexible printed circuit board **1** has a planar shape of a rectangle in FIG. **1**A, the planar shape of the flexible printed circuit board **1** is not limited to a shape of the rectangle. For example, the width of the flexible printed circuit board **1** maybe partially changed in the length direction, a protrusion may be formed near the electrode terminal of the flexible printed circuit board **1**, or the flexible printed circuit board **1** may have a slit in it.
(13) Further, although not illustrated, a single layer or multiple layers of a conductive shielding layer, a protective film, a reinforcing film, and the like may be formed on both or either one of the flexible base **2** and the coverlay **4** of the flexible printed circuit board **1**. These layers may be formed so as to cover the entire surface of the flexible printed circuit board **1**, or may be formed so as to cover a part of the flexible printed circuit board **1**.
(14) The flexible base **2** is a sheet-like or film-like insulating base made of a resin or the like, and has plasticity and flexibility. Therefore, the flexible printed circuit board **1** can be deformed such as bending or curving. The insulator forming the flexible base **2** may have electrical insulating property. For example, polyimide, polyethylene terephthalate, or the like is used as the insulator forming the flexible base **2**.
(15) The flexible wiring layer **3** is a conductive layer made of copper foil or other metal foil. The flexible wiring layer **3** has wiring patterns. The flexible wiring layer **3** is formed on one surface or both surfaces of the flexible base **2**. The conductor forming the flexible wiring layer **3** is a material having higher conductivity and thermal conductivity than the insulator, and is a metal such as copper, silver, gold, or the like. The flexible wiring layer **3** maybe formed on at least one surface of the flexible base **2**.
(16) The coverlay **4** is an insulating layer for protecting a circuit formed by the flexible wiring layer **3**. The coverlay **4** is formed of a cover film, an overcoat, or the like. The coverlay **4** is formed on the surface of the flexible base **2** on which the flexible wiring layer **3** is formed so as to cover the flexible wiring layer **3**.
(17) The coverlay **4** is not formed at one end of the flexible printed circuit board **1**, and the flexible wiring layer **3** is exposed. A first electrode **71** is formed on an exposed portion of the flexible wiring layer **3**. Further, gold, tin, or the like may be plated on the first electrode **71**. A plurality of first electrodes **71** are arranged side by side at a predetermined pitch in the X direction. The plurality of first electrodes **71** form a first electrode group. Each of the plurality of first electrodes **71** has a planar shape of a rectangle having a short side along the X direction and a long side along the Y direction. All the first electrodes **71** may be arranged at the same pitch or at different pitches. Thus, the first electrode **71** is formed by the flexible wiring layer **3** exposed at the end portion of the flexible printed circuit board **1**. Although not illustrated, an electrode may be formed in the same manner at the other end of the flexible printed circuit board **1**, and the other end portion may be an inserting terminal. Further, on the surface of the coverlay **4**, the flexible wiring layer **3** maybe partially exposed as an electrode for mounting a connector component in the form of surface

mounting.

(18) The printed wiring board **10** includes a printed wiring base **11**, a plurality of conductive layers **12**, **13**, **14**, and **15** as a plurality of wiring layers, and a solder resist layer **17**. The printed wiring board **10** is configured such that the plurality of conductive layers **12**, **13**, **14**, and **15** are stacked via the printed wiring base **11**. Unlike the flexible printed circuit board **1**, the printed wiring board **10** is a rigid wiring board. For example, the printed wiring board **10** maybe formed of a glass epoxy material or a ceramic substrate. Further, instead of the printed wiring board **10**, a printed wiring board in which elements such as an imaging element are arranged on a ceramic substrate and the ceramic substrate and the printed wiring board are connected to each other by a pair of electrodes via solder can be used.

(19) The printed wiring base **11** is an insulating base made of a rigid composite material or the like and shaped in a board, for example. Unlike the flexible base **2**, the printed wiring base **11** is a rigid material. The insulator forming the printed wiring base **11** may have electrical insulation. For example, as the printed wiring base **11**, a resin substrate obtained by curing a resin such as epoxy resin, a ceramic substrate using ceramic, or the like is used.

(20) The conductive layers **12**, **13**, **14**, and **15** as the wiring layers are conductive layers made of copper foil or other metal foil. In the printed wiring board **10**, the wiring layers have wiring patterns. The wiring layer is formed on one surface or both surfaces of the printed wiring base **11**. One or multiple wiring layers are also formed inside the printed wiring base **11**.

(21) FIG. **1**C illustrates a case where the two conductive layers **12** and **15** are formed on both surfaces of the printed wiring base **11**, the two conductive layers **13** and **14** are formed inside, and thus a total of four wiring layers are formed. In the present embodiment, the case where the wiring layers in the printed wiring board **10** are four layers will be described, but the present invention is not limited to four layers. The wiring layer in the printed wiring board **10** maybe a single layer or multiple layers, that is, may be four or less layers or four or more layers.

(22) The conductive layer **12** is formed on the surface layer of the printed wiring board **10**, that is, on one surface of the printed wiring base **11**. The conductive layer **12** is a first conductive layer to which the flexible printed circuit board **1** is connected via the solder **5**. The conductive layer **13** is a second conductive layer formed under the first conductive layer **12** inside the printed wiring base **11** from the conductive layer **12**. In the following description, the conductive layer **13** is referred to as a “second conductive layer **13**” as appropriate. The conductive layer **14** is a third conductive layer formed in a second layer inside the printed wiring base **11** from the conductive layer **12**. The conductive layer **15** is a fourth conductive layer formed on the surface layer of the printed wiring board **10**, that is, on the other surface of the printed wiring base **11**. Vias electrically connecting the conductive layers **12**, **13**, **14**, and **15** are formed inside the printed wiring base **11**. FIG. **1**C illustrates a via **16** electrically connecting the conductive layer **13** and the conductive layer **14**, but vias electrically connecting the other conductive layers are also appropriately formed. The conductor of the conductive layers **12**, **13**, **14**, and **15** and the via **16** is a material having higher conductivity and thermal conductivity than the insulator, and is, for example, a metal such as copper, gold, or the like.

(23) The solder resist layer **17** is an insulating protective film for protecting a circuit formed by the conductive layers **12** and **15** as the wiring layers. The solder resist layer **17** is formed of a cured liquid resist, a film-like solder resist, or the like. The solder resist layer **17** is formed on the one surface and the other surface of the printed wiring base **11** so as to cover the conductive layers **12** and **15**.

(24) Openings through which the conductive layers **12** and **15** are exposed are formed in the solder resist layer **17** formed on the both surfaces of the printed wiring base **11**. On the one surface of the printed wiring base **11**, the exposed portion of the conductive layer **12** forms second electrodes **72** and third electrodes **73**. That is, the conductive layer **12** is formed to include the second electrodes **72** and the third electrodes **73**.

(25) A plurality of the second electrodes **72** are arranged, for example, at the center of the printed wiring board **10**. The plurality of second electrodes **72** form a second electrode group. The second electrode **72** is electrically connected to the first electrode **71** of the flexible printed circuit board **1** by soldering via the solder **5**. In this manner, in the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10**, the first electrode group formed of the plurality of first electrodes **71** and the second electrode group formed of the plurality of second electrodes **72** are connected by soldering via the solder **5**.
(26) The third electrodes **73** are arranged around the second electrodes **72**, for example. The third electrode **73** is connected to the peripheral component **40** via the solder **5**.
(27) The peripheral component **40** is an electronic component mounted on the printed wiring board **10**, and is, for example, an active component such as an IC or a crystal (crystal resonator) which is relatively susceptible to heat. The peripheral component **40** may be a passive component such as a chip capacitor, a chip resistor, or the like. The peripheral component **40** includes fourth electrodes **74**. The third electrodes **73** are electrically connected to the fourth electrodes **74** of the peripheral component **40** by soldering via the solder **5**.
(28) In the case where the first electrode **71** and the second electrode **72** are connected by using the solder **5**, the first electrode **71** and the second electrode **72** can be bonded to the solder **5** in a state where the solder **5** is heated to a melting point or higher. The solder **5** maybe, for example, Sn-3.0Ag-0.5Cu solder or Sn-58Bi solder pasted together with flux. The same applies to the case where the fourth electrode **74** and the third electrode **73** are connected by using the solder **5**.
(29) FIG. **1**B illustrates a reference region **50** relating to the connected portion **6** which is a bonded portion between the flexible printed circuit board **1** and the printed wiring board **10**, and first to third regions **51**, **52**, and **53** in a plane including the second conductive layer **13**. The reference region **50** and the first to third regions **51**, **52**, and **53** are regions in a plan view as viewed in the Z direction, that is, regions in a plan view with respect to a plane including the second conductive layer **13**. Here, the reference region **50**, the first to third regions **51**, **52**, **53**, and the like in the second conductive layer **13** when viewed in a plan view in the Z direction will be described. The units of the lengths W**1**, W**2**, W**2**, L**1**, L**2**, and L**3** in the following description are, for example, mm.
(30) The reference region **50** relating to the bonded portion between the flexible printed circuit board **1** and the printed wiring board **10** is a region relating to the connected portion of the first electrodes **71** of the flexible printed circuit board **1** and the second electrodes **72** of the printed wiring board **10** by soldering via the solder **5**. The reference region **50** has a contour of a rectangle, and is set such that four sides of the rectangle are circumscribed to the bonded portion between the first electrode group including the plurality of first electrodes **71** and the second electrode group including the plurality of second electrodes **72** in a plan view with respect to a plane including the second conductive layer **13**. Specifically, as illustrated in FIG. **1**B, the reference region **50** is a region surrounded by both left and right ends of the plurality of first electrodes **71** in the arrangement direction of the plurality of first electrodes **71** in the X direction, which is the width direction of the flexible printed circuit board **1**, and both upper and lower ends in the Y direction, which is the longitudinal direction of the flexible printed circuit board **1**. The reference region **50** has a planar shape of a rectangle in which the length of the long side along the X direction is W**1** and the length of the short side along the Y direction is L**1**. Here, the X direction is a direction in which the plurality of first electrodes **71** in the flexible printed circuit board **1** are arranged. The Y direction is a direction in which the wirings of the flexible wiring layer **3** and the first electrodes **71** which are parts of the wirings in the flexible printed circuit board **1** extend.
(31) The short side **60** of the reference region **50** is a side having a length L**1** in the long side direction along the Y direction of the solder **5** bonding the first electrode **71** of the flexible printed circuit board **1** and the second electrode **72** of the printed wiring board **10**. The long side **63** of the reference region **50** is a side having a length W**1** from an electrode end at one end to an electrode

end at the other end in the X direction in which the first electrodes **71** of the flexible printed circuit board **1** are arranged.

(32) The first region **51** of the plane including the second conductive layer **13** of the printed wiring board **10** is a region having a shape of a rectangle and including the reference region **50**, which is wider than the reference region **50**. The long side of the first region **51** along the X direction has a length of W**1**+2×W**2**. The short side of the first region **51** along the Y direction has a length of L**1**+2×L**2**. Here, the length L**2** is the length of the distance **61** from the boundary of the reference region **50** in the Y direction to the boundary of the first region **51** in the Y direction. The length W**2** is the length of the distance **64** from the boundary of the reference region **50** in the X direction to the boundary of the first region **51** in the X direction. A region in which the reference region **50** is projected onto the second conductive layer **13** of the printed wiring board **10** in the −Z direction is included in the first region **51**.

(33) At least one of the length L**2** and the length W**2** is preferably 0.5 times or more the length L**1**, and more preferably 1 times or more the length L**1**. At least one of the length L**2** and the length W**2** is preferably 5 times or less the length L**1**, preferably 4 times or less the length L**1**, and preferably 3.5 times or less the length L**1**. At least one of the length L**2** and the length W**2** is preferably 3 times or less the length L**1**, preferably 2.5 times or less the length L**1**, and preferably 2 times or less the length L**1**. At least one of the length L**2** and the length W**2** may be 1 times or more and 3 times or less the length L**1**, 1 times or more and 2 times or less the length L**1**, or 1 times or more and 1.5 times or less the length L**1**. Further, the length L**2** and the length W**2** may be equal to each other, and when the length L**2** and the length W**2** are different from each other, either one of the two may be long.

(34) The second region **52** of the plane including the second conductive layer **13** of the printed wiring board **10** is located around the outside of the first region **51** and has an annular planar shape of a rectangle that does not include the first region **51**. The second region **52** is in contact with the first region **51** through a first boundary B**1**. The long side of the outer shape of the second region **52** along the X direction has a length of W**1**+2×W**2**+2×W**3**. The short side of the outer shape of the second region **52** along the Y direction has a length of L**1**+2×L**2**+2×L**3**. Here, the length L**3** is the length of the distance **62** from the boundary in the Y direction of the first region **51** to the boundary in the Y direction of the second region **52**. The length W**3** is the length of the distance **65** from the boundary in the X direction of the first region **51** to the boundary in the Y direction of the second region **52**. In a plan view with respect to a plane including the second conductive layer **13**, that is, in a plan view as viewed in the Z direction, the second region **52** can overlap the flexible printed circuit board **1** depending on the position of the connected portion **6** in the printed wiring board **10**.

(35) At least one of the length L**3** and the length W**3** is preferably 0.5 times or more the length L**1**, preferably 1 times or more the length L**1**, preferably 1.5 times or more the length L**1**. At least one of the length L**3** and the length W**3** is preferably 5 times or less the length L**1**, preferably 4 times or less the length L**1**, and preferably 3.5 times or less the length L**1**. At least one of the length L**3** and the length W**3** is preferably 3 times or less the length L**1**, and may be 2.5 times or less the length L**1**. At least one of the length L**3** and the length W**3** is more preferably 1.5 times or more and 2.5 times or less the length L**1**. At least one of the length L**3** and the length W**3** may be 2 times or 3 times the length L**1**. The lengths L**3** and W**3** may be equal to each other, and when the lengths L**2** and W**2** are different, either one of the two may be longer.

(36) A plurality of openings **20** are periodically formed in the second conductive layer **13** in the second region **52**. Each opening **20** is formed so as to penetrate the second conductive layer **13** in the Z direction. The planar shape of the opening **20** as viewed in the Z direction is, for example, a circle. For example, as illustrated in FIG. **1**B, the plurality of openings **20** are formed in such a manner that the positions in the Y direction are arranged in a staggered manner in three lines along the Y direction in each of the one band portion and the other band portion along the Y direction of the second region **52**. Further, the plurality of openings **20** are formed in such a manner that the

positions in the Y direction are arranged in a staggered manner in two lines along the X direction in each of the one band portion and the other band portion along the X direction of the second region **52**.
(37) The number of lines of the openings **20** in each band portion of the second region **52** is not limited to two or three, and may be two or more columns, or may be one. Further, the arrangement of the openings **20** does not necessarily have to be staggered, and may be periodically arranged or non-periodically arranged, and may be arranged at the same pitch or at different pitch. The planar shape of the opening **20** when viewed in the Z direction is not limited to a circle, and may be a polygon such as a rectangle, an ellipse, a shape having a concave portion or a convex portion, or the like. In addition, one or a plurality of openings **20** can be formed in the second conductive layer **13** of the second region **52** according to the required heat dissipation performance and the like, and a predetermined number of openings **20** can be formed.
(38) The third region **53** of the plane including the second conductive layer **13** of the printed wiring board **10** is a region excluding the first region **51** and the second region **52** from the plane including the second conductive layer **13**. The third region **53** is positioned around the outside of the second region **52**. The third region **53** is in contact with the second region **52** through a second boundary B**2**.
(39) The first boundary B**1** and the second boundary B**2**, which are the inner and outer boundaries of the second region **52** arranged as described above, are positioned so as to satisfy the following relationship. Note that the first to fourth directions in the following description are one of the four directions of the +X direction, the −X direction, the +Y direction, and the −Y direction, and are different directions.
(40) First, the first boundary portion, which is at least a part of the first boundary B**1**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the first direction from at least the first side, which is one side of the four sides of the reference region **50**, toward the outside of the reference region **50**. Preferably, the first boundary portion is positioned within a range of 1 times or more and 2 times or less the length L**1** of the short side **60** in the first direction from at least the first side, which is one of the four sides of the reference region **50**, toward the outside of the reference region **50**. The second boundary portion, which is at least a part of the second boundary B**2**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the first direction from the first boundary portion of the first boundary B**1**. Preferably, the second boundary portion is positioned within a range of 1.5 times or more and 2.5 times or less the length L**1** of the short side **60** in the first direction from the first boundary portion of the first boundary B**1**.
(41) The third boundary portion, which is at least a part of the first boundary B**1**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the second direction from at least the second side, which is one side of the four sides of the reference region **50**, toward the outside of the reference region **50**. Preferably, the third boundary portion is positioned within a range of 1 times or more and 2 times or less the length L**1** of the short side **60** in the second direction from at least the second side, which is one of the four sides of the reference region **50**, toward the outside of the reference region **50**. The fourth boundary portion, which is at least a part of the second boundary B**2**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the second direction from the third boundary portion of the first boundary B**1**. Preferably, the fourth boundary portion is positioned within a range of 1.5 times or more and 2.5 times or less the length L**1** of the short side **60** in the second direction from the third boundary portion of the first boundary B**1**.
(42) The fifth boundary portion, which is at least a part of the first boundary B**1**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the third direction from at least the third side, which is one side of the four sides of the reference region **50**, toward the outside of the reference region **50**. Preferably, the fifth boundary portion is

positioned within a range of 1 times or more and 2 times or less the length L**1** of the short side **60** in the third direction from the at least third side, which is one of the four sides of the reference region **50**, toward the outside of the reference region **50**. The sixth boundary portion, which is at least a part of the second boundary B**2**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the third direction from the fifth boundary portion of the first boundary B**1**. Preferably, the sixth boundary portion is positioned within a range of 1.5 times or more and 2.5 times or less the length L**1** of the short side **60** in the third direction from the fifth boundary portion of the first boundary B**1**.

(43) The seventh boundary portion, which is at least a part of the first boundary B**1**, is positioned within a range of 0.5 times or more and 5 times or less the length L**1** of the short side **60** in the fourth direction from at least the fourth side, which is one side of the four sides of the reference region **50**, toward the outside of the reference region **50**. Preferably, the seventh boundary portion is positioned within a range of 1 times or more and 2 times or less the length L**1** of the short side **60** in the fourth direction from at least the fourth side, which is one of the four sides of the reference region **50**, toward the outside of the reference region **50**. The eighth boundary portion, which is at least a part of the second boundary B**2**, is positioned within a range of 0.5 times or more and 5 times or less of the length L**1** of the short side **60** in the fourth direction from the seventh boundary portion of the first boundary B**1**. Preferably, the eighth boundary portion is positioned within a range of 1.5 times to 2.5 times the length L**1** of the short side **60** in the fourth direction from the seventh boundary portion of the first boundary B**1**.

(44) The above relationship may be satisfied for at least one of the first to fourth sides which are the four sides of the reference region **50**. In addition, it is preferable that one side of the four sides of the reference region **50** that satisfy the above relationship is the long side of the four sides. The four sides of the reference region **50** include two pairs of opposite sides.

(45) The density of the second conductive layer **13** in the first region **51** is higher than the density of the second conductive layer **13** in the second region **52**. The first region **51** has a relatively high density of the second conductive layer **13** and has, in addition to the region in which the reference region **50** is projected onto the second conductive layer **13**, around the region, an annular region of the rectangle, in which the width in the X direction is W**2** and the width in the Y direction is L**2**. Therefore, in the first region **51**, the heat from the connected portion **6** due to the solder **5** between the flexible printed circuit board **1** and the printed wiring board **10** can be diffused in the planar direction, and the heat dissipation property can be improved.

(46) From the viewpoint of further improving the heat dissipation property, it is preferable that the density of the second conductive layer **13** in the region excluding the reference region **50** from the first region **51** in the plan view when viewed in the Z direction is higher than the density of the second conductive layer **13** in the reference region **50**. Here, the plan view when viewed in the Z direction is the plan view with respect to the plane including the second conductive layer **13**. By having such a density difference in the second conductive layer **13**, heat dissipation from the second conductive layer **13** can be promoted and the heat dissipation property can be further improved. In addition, the second conductive layer **13** in the region excluding the region overlapping with the reference region **50** from the first region **51** is preferably a solid pattern. This pattern can also further improve heat dissipation.

(47) Further, from the viewpoint of ensuring heat dissipation, the density of the second conductive layer **13** in the first region **51** is preferably 50% or more in area ratio.

(48) On the other hand, in the second region **52** arranged around the first region **51**, the density of the second conductive layer **13** is lower than that of the first region **51**. Therefore, in the second region **52**, it is possible to suppress transfer of heat from the connected portion **6** which is a bonded portion between the flexible printed circuit board **1** and the printed wiring board **10** by the solder **5** to the outside of the second region **52** in the inner layer of the printed wiring board **10** to a low level. Further, since the plurality of openings **20** are formed in the second region **52**, heat transfer

can be effectively suppressed to a low level. In this way, the second region **52** can suppress the influence of heat of the solder **5** between the flexible printed circuit board **1** and the printed wiring board **10** on the periphery of the connected portion **6** in the inner layer of the printed wiring board **10**. This makes it possible to suppress the influence of heat on the peripheral components **40** mounted on the printed wiring board **10**, in particular, the influence of heat on the peripheral component **40**, which is an active component adversely affected by heat.
(49) In the present embodiment, the case where the reference region **50** is positioned at the center of the first region **51** in the plan view viewed in the Z direction has been described, but the relationship between the first region **51** and the reference region **50** is not limited thereto. The first region **51** may include the reference region **50** in the plan view as viewed in the Z direction, and may be a region positioned to the outside of the reference region **50** in at least one direction. In this case, the reference region **50** may be located not only at the center of the first region **51** as in the present embodiment but also inside the first region **51**. The planar shape of the first region **51** is not limited to the rectangle, and may be a polygon other than a rectangle, a shape including a curve such as a circle or an ellipse, or a shape including concave and convex. In this case, the first region **51** may be a region having the length that is 1 times or more and 2 times or less the length L**1** of the short side **60** of the reference region **50** outside the reference region **50** in the Y direction, which is the direction along the short side **60** of the reference region **50**, in the plan view as viewed in the Z direction.
(50) Further, in the present embodiment, the case where the second region **52** is positioned around the first in the plan view viewed in the Z direction has been described, but the relationship between the first region **51** and the second region **52** is not limited thereto. The second region **52** may be a region located outside the first region **51** in the plan view as viewed in the Z direction. Further, the planar shape of the second region **52** is not limited to an annular shape of the rectangle, and may be an annular shape having an outer shape of a polygon other than a rectangle, a shape including a curved line such as, a circle, an ellipse, or the like, or a shape including concave and convex, or the like, or a shape in which a part of the annular shape is missing. In this case, the second region **52** may be a region having a length that is 1.5 times or more and 2.5 times or less the length L**1** of the short side **60** of the reference region **50** outside the first region **51** in the X direction which is the direction along the short side **60** of the reference region **50** in the plan view as viewed in the Z direction.
(51) The peripheral component **40** includes fourth electrodes **74**. The fourth electrodes **74** of the peripheral component **40** are electrically connected to the third electrodes **73** of the printed wiring board **10** by soldering via the solder **5**.
(52) The arrangements of the peripheral components **40** will now be described with reference to FIG. **1**A. FIG. **1**A illustrates a boundary line **80** in which the boundary of the first region **51** is projected in the +Z direction onto the upper surface of the printed wiring board **10**, and a boundary line **81** in which the boundary of the second region **52** is projected in the +Z direction onto the upper surface of the printed wiring board **10**.
(53) The peripheral component **40**, which is an active component, is preferably arranged at a position outside the boundary line **81** from the viewpoint of suppressing the above-described influence of heat. That is, the peripheral component **40**, which is the active component, is preferably arranged on the third region **53**.
(54) On the other hand, when the peripheral component **40** is a passive component such as a chip capacitor, a chip resistor, or the like, the passive component may have a higher heat resistance than the peripheral component **40**, such as the active component which may be arranged on the third region **53**. If the peripheral component **40** has a higher heat resistance than the peripheral component **40** that may be arranged on the third region **53**, such as the passive component, the peripheral component **40** may be arranged at a position outside the boundary line **80** and inside the boundary line **81**. That is, the peripheral component **40** having higher heat resistance than the

peripheral component **40** which may be arranged on the third region **53** may be arranged on the second region **52**.
(55) The peripheral components **40** are not limited to be arranged in the above-described arrangement, and may be arranged at appropriate positions according to the design of the electronic module **100** or the like and mounted on the printed wiring board **10**.
(56) Although not illustrated, the minimum component necessary for the operation of the electronic module **100** are mounted on the printed wiring board **10**. The flexible base **2** maybe covered with, for example, an ultraviolet (UV) curing resin, a thermosetting resin, a film-like reinforcing part, or the like only at both ends in the X direction, which is the width direction, or over the X direction, which is the width direction. Thereby, the structure of the connected portion **6** by the solder **5** and the reinforcing material improve the connection strength.
(57) Although FIGS. **1**A to **1**C illustrate a case where the flexible printed circuit board **1** is connected to the printed wiring board **10** such that the longitudinal direction of the flexible printed circuit board **1** is parallel to the short side of the printed wiring board **10**, the present invention is not limited thereto. The flexible printed circuit board **1** maybe connected to the printed wiring board **10** such that the longitudinal direction of the flexible printed circuit board **1** is parallel to the long side of the printed wiring board **10**.
(58) Although FIGS. **1**A to **1**C illustrate a case where one flexible printed circuit board **1** is connected to the printed wiring board **10**, the present invention is not limited thereto. The plurality of flexible printed circuit boards **1** maybe connected to the printed wiring board **10**. When the plurality of flexible printed circuit boards **1** are connected to the printed wiring board **10**, the directions in which the plurality of flexible printed circuit boards **1** are leaded out may be the same direction or different directions.
Second Embodiment
(59) Next, an electronic module according to a second embodiment will be described with reference to FIG. **2**. FIG. **2** is the plan view illustrating the second conductive layer **13** of the second layer from the surface layer when the electronic module **100** according to the present embodiment is viewed from the top.
(60) In this embodiment, the shape of the opening **20** in the second region **52** is different. That is, in the present embodiment, as illustrated in FIG. **2**, the second conductive layer **13** in the second region **52** is formed with a plurality of openings **21** instead of a part of the plurality of openings **20**. Each opening **21** is formed so as to penetrate the second conductive layer **13** in the Z direction. The opening **21** has an elliptical planar shape as the planar shape when viewed in the Z direction, and has a larger opening area than the opening **20**. The plurality of openings **21** are formed at the centers of one and the other of the band portions along the X direction of the second region **52** positioned along the long side of the first region **51**. That is, the opening **21** is formed at a position closer to the center of the long side **63** than the end of the long side **63** of the reference region **50**. The central portion of the long side **63** means two middle portions of the long side **63** divided into four equal portions, and the peripheral portion of the long side **63** means 25% of each of both end portions (left and right) of the long side **63**. Thus, in the present embodiment, the opening **21** serving as a first opening and the opening **20** serving as a second opening whose opening area is smaller than that of the opening **21** are formed in the second conductive layer **13** in the second region **52**. The other configurations are the same as those of the first embodiment.
(61) As described above, in the present embodiment, the opening **21** formed in the central portion of the band portion along the X direction of the second region **52** positioned along the long side of the first region **51** has the elliptic planar shape, and has the larger opening area than the peripheral opening **20**. Thus, the opening area of the opening **21** adjacent to the center of the reference region **50** is larger than those of the other openings **20**. As in the case of the flexible printed circuit board **1**, a rectangular connected portion is formed as the connected portion **6** with the printed wiring board **10**, and when the connected portion **6** generates heat, the heat generation portions thermally

interfere with each other, and as a result, the temperature of the rectangular center portion becomes higher than that of the surroundings. Therefore, in the case where all the openings **20** arranged in the second region **52** have the same shape, it may be difficult to suppress the influence of heat caused by the heat in the central portion to be small. On the other hand, in the present embodiment, since the opening area of the central portion of the second region **52** is selectively increased by the elliptical opening **21**, even when the central portion of the connected portion **6** generates more heat, it is possible to suppress the influence of heat on the surroundings to be small.

(62) The planar shape of the opening **21** when viewed in the Z direction is not limited to shape of ellipse, and may be a polygon such as a rectangle, a shape of circle, a shape having a concave portion or a convex portion, or the like. In addition, one or the plurality of openings **21** can be formed in the second conductive layer **13** of the second region **52** according to the required heat dissipation performance and the like, and a predetermined number of openings **21** can be formed. The opening **20** maybe formed such that the second conductive layer **13** is continuous from the opening **20** to the first region **51**. Also, the opening **21** can be formed so that the second conductive layer **13** is continuous from the opening **21** to the first region **51**.

Third Embodiment

(63) Next, an electronic module **100** according to a third embodiment will be described with reference to FIG. **3**. FIG. **3** is the top plan view of the electronic module **100** according to the present embodiment.

(64) In the present embodiment, as illustrated in FIG. **3**, a heat dissipation part **31** as the heat dissipation part is arranged at a connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10**. The heat dissipation part **31** is formed on the upper surface of the printed wiring board **10** so as to cover the connected portion **6** which is a bonded portion between the flexible printed circuit board **1** and the printed wiring board **10**. The flexible printed circuit board **1** is provided between the heat dissipation part **31** and the printed wiring board **10** so that heat from the connected portion **6** is transferred to the heat dissipation part **31** through the flexible printed circuit board **1**. Further, in the present embodiment, the flexible printed circuit board **1** has protruding portions **32**. The protruding portions **32** protrude outward in the X direction from both side portions of the flexible printed circuit board **1** along the Y direction. The protruding portion **32** is positioned on the printed wiring board **10**. The protruding portion **32** is reinforced by the reinforcing part **30** and fixed to the printed wiring board **10**. The other configurations are the same as those of the first embodiment.

(65) As described above, in the present embodiment, the heat dissipation part **31** is arranged so as to cover the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10**. Although the material of the heat dissipation part **31** is not particularly limited, the heat dissipation part **31** is made of a material having excellent thermal conductivity, such as a metal, ceramics, a carbon member such as a graphite sheet, or the like. Although the shape of the heat dissipation part **31** is not particularly limited, the heat dissipation part **31** has, for example, the shape of a rectangle having the long side along the X direction and the short side along the Y direction.

(66) It is preferable that the heat dissipation part **31** is arranged such that the short side thereof straddles the boundary line **80** and is positioned inside the boundary line **81**. As described above, the boundary line **80** is a line in which the boundary of the first region **51** is projected onto the upper surface of the printed wiring board **10** in the +Z direction, and the boundary line **81** is a line in which the boundary of the second region **52** is projected onto the upper surface of the printed wiring board **10** in the +Z direction. By arranging the heat dissipation part **31** in this manner, the heat of the connected portion **6** of the flexible printed circuit board **1** can be dissipated more efficiently. Further, since the heat dissipation part **31** is arranged inside the boundary line **81**, heat conducted to the second region **52** of the printed wiring board can also be absorbed by the heat dissipation part **31**, so that the influence of heat on the peripheral component **40**, which is an active

component, can be suppressed to a small extent.
(67) In addition, the protruding portions **32** formed along the X direction, which is the width direction of the flexible printed circuit board **1**, are fixed to the printed wiring board **10** by the reinforcing parts **30**. The reinforcing part **30** is made of, for example, a resin such as an ultraviolet curing resin. Similarly to the heat dissipation part **31**, since the reinforcing part **30** absorbs heat, it is possible to suppress the influence of heat on the peripheral component **40**. Further, in addition to the reduction of the influence of heat by the reinforcing part **30**, the strength of the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10** can be improved by the reinforcing parts **30**.
(68) In the present embodiment, the electronic module **100** is configured to include the reinforcing parts **30**, the heat dissipation part **31**, and the protruding portions **32** in addition to the configuration of the first embodiment. In addition to the configuration of the second embodiment, the electronic module **100** may be configured to include the reinforcing parts **30**, the heat dissipation part **31**, and the protruding portions **32** as in the present embodiment.
Fourth Embodiment
(69) Next, an electronic module **100** according to a fourth embodiment will be described with reference to FIGS. **4**A to **4**C. FIG. **4**A is the plan view illustrating the second conductive layer **13** of the second layer from the surface layer when the electronic module **100** according to the present embodiment is viewed from the top. FIG. **4**B is an enlarged plan view of an opening **90** formed in the first region **51**. FIG. **4**C is a cross-sectional view taken along the line B-B′ in FIG. **4**B.
(70) In the present embodiment, as illustrated in FIG. **4**A, in addition to the configuration of the second embodiment, rectangular openings **90** are formed as third openings in the second conductive layer **13** in the first region **51**. Other configurations are similar to those of the second embodiment. In the present embodiment, since the rectangular openings **90** are formed in the second conductive layer **13** of the first region **51**, heat conduction in the surface direction can be suppressed while heat dissipation in the first region **51** is maintained.
(71) In FIG. **4**B, one of the rectangular openings **90** formed in the second conductive layer **13** in the first region **51** is enlarged, and the second electrode **72** in the conductive layer **12** which is the first conductive layer of the printed wiring board **10** is illustrated superimposed in the thickness direction of the printed wiring board **10** which is the −Z direction. The rectangular opening **90** has a planar shape of a rectangle having a short side **91** of a length L**5** along the X direction and a long side **92** of a length L**4** along the Y direction. The rectangular opening **90** includes, for example, two rectangular terminals of the second electrodes **72** of the printed wiring board **10** in the X direction, which is the arrangement direction of the rectangular terminals in the short side **91**. That is, the rectangular opening **90** is formed so as to overlap with, for example, the second electrodes **72** of two terminals in a plan view when viewed in the Z direction. Note that the number of terminals of the second electrodes **72** where the rectangular opening **90** overlaps is not limited to two, and may be one terminal, or may be three or more terminals.
(72) FIG. **4**C is a cross-sectional view taken along the line B-B′ in FIG. **4**B. The rectangular opening **90** extends through the second conductive layer **13** in the Z direction. One end **93** of the long side **92** of the rectangular opening **90** of the first region **51** coincides with one end **94** of the long side of the bonded portion formed by the solder **5**. That is, the second electrode **72** and the rectangular opening **90** overlap each other in the plan view as viewed in the Z direction, which is the thickness direction of the printed wiring board **10**.
(73) Further, each terminal of the second electrode **72** and the first electrode **71** connected to the second electrode **72** located above the rectangular opening **90** may be a signal terminal through which a signal is transmitted. In the case of the signal terminal, the impedance characteristic can be improved by the opening **90**, and the high-speed signal can be supported. As described above, in the present embodiment, it is possible to achieve both the improvement of the signal characteristics and the improvement of the heat dissipation characteristics. From the viewpoint of achieving both

of these points, the length L**5** of the short side **91** of the rectangular opening **90** is preferably 1 times or more and 3 times or less the length of the short side of the second electrode **72** and the first electrode **71** connected to the second electrode **72** in the X direction, which is the direction along the short side **91**. From the same viewpoint, the length L**4** of the long side **92** of the rectangular opening **90** is more preferably 1 times or more and 2 times or less the length of the long side that is the long side of the second electrode **72** and the first electrode **71** connected to the second electrode **72** in the Y direction that is the direction along the long side **92**. The opening **90** need not necessarily have a planar shape of the rectangle, and may have a planar shape having a short side and a long side. In this case, the length of the short side of the opening **90** corresponding to the short side **91** is preferably 1 times or more and 3 times or less the length of the short side of the second electrode **72** and the first electrode **71** connected to the second electrode **72** in the direction along the short side. The length of the opening **90** in the longitudinal direction corresponding to the long side **92** is more preferably 1 times or more and 2 times or less the length of the second electrode **72** and the first electrode **71** connected to the second electrode **72** in the longitudinal direction.

(74) In the present embodiment, in addition to the configuration of the second embodiment, the case where the opening **90** are formed in the first region **51** has been described, but the present invention is not limited thereto. In addition to the configuration of the first or third embodiment, the openings **90** may be formed in the first region **51** as in the present embodiment.

Fifth Embodiment

(75) Next, an imaging unit **400** according to a fifth embodiment will be described with reference to FIGS. **5**A and **5**B. FIG. **5**A is a schematic top view illustrating a schematic configuration of the imaging unit **400** according to the present embodiment. FIG. **5**B is a schematic cross-sectional view illustrating a schematic configuration of the imaging unit **400** according to the present embodiment, and illustrates a cross-sectional view taken along a line C-C′ in FIG. **5**A. In the present embodiment, an imaging unit **400** configured using the electronic module **100** according to any one of the first to fourth embodiments will be described.

(76) As illustrated in FIGS. **5**A and **5**B, the imaging unit **400** includes the electronic module **100** according to any one of the first to fourth embodiments, an imaging element **27**, a frame **25**, and a cover glass **26**. The electronic module **100** includes the printed wiring board **10** and the flexible printed circuit board **1**.

(77) The frame **25** is adhered and fixed by an adhesive such as an ultraviolet curing resin to a peripheral end portion of a surface of the printed wiring board **10** opposite to a surface to which the flexible printed circuit board **1** is connected. The cover glass **26** is attached to the frame **25** so as to face the printed wiring board **10**. The imaging element **27** is mounted on a surface surrounded by the frame **25** and the cover glass **26** of the printed wiring board **10**. The surface of the printed wiring board **10** on which the imaging element **27** is mounted is not necessarily the surface opposite to the surface to which the flexible printed circuit board **1** is connected, and may be the surface to which the flexible printed circuit board **1** is connected.

(78) The imaging unit **400** is held by the camera shake correction unit **410** so as to be movable with respect to the camera shake correction unit **410**. Although not illustrated, the imaging unit **400** is connected to a connector mounted on the printed wiring board of an image processing unit via the other inserting terminal of the flexible printed circuit board **1** or a connector component mounted on the flexible printed circuit board **1**. In this way, the flexible printed circuit board **1** electrically connects the imaging unit **400** and the image processing unit to each other.

(79) The imaging element **27** is a solid-state imaging element such as a CMOS (Complementary Metal Oxide Semiconductor) image sensor, a CCD (Charge Coupled Device) image sensor, or the like. The imaging element **27** is attached to the printed wiring board **10** so as not to contact the cover glass **26** in a hollow portion surrounded by the printed wiring board **10**, the cover glass **26**, and the frame **25**. The frame **25** maybe a metal frame or a resin frame. The imaging element **27** is

electrically connected to wire pads **29** of the printed wiring board **10** through metal wires **28**. The wire pad **29** is, for example, gold-plated. For example, the imaging element **27** of an LGA (Land Grid Array) type or a CLCC (Ceramic Leadless Chip Carrier) type can be used.
(80) The imaging unit **400** is attached to and fixed to a frame **401** which is, for example, a metal frame. The frame **401** is fixed to the peripheral portion of the surface of the printed wiring board **10** to which the flexible printed circuit board **1** is connected in the imaging unit **400**. The camera shake correction unit **410** supports the frame **401** so that the imaging unit **400** fixed to the frame **401** can be moved in the X direction and the Y direction and can be rotated in a rotation direction around a rotation axis along the Z axis. The camera shake correction unit **410** can correct the camera shake by moving or rotating the imaging unit **400** in accordance with the camera shake.
(81) In the present embodiment, the case where the frame **25** is attached is described, but the position where the frame **25** is arranged is not limited to the peripheral portion of the printed wiring board **10**. The printed wiring board **10** maybe embedded in the cavity-shaped frame **25**. The imaging element **27** maybe arranged in a hollow portion of the printed wiring board **10** having a counterbore, such as a cavity substrate.
Sixth Embodiment
(82) Next, electronic equipment according to a sixth embodiment will be described with reference to FIG. **6**. FIG. **6** is a schematic diagram illustrating a schematic configuration of an imaging apparatus as an example of the electronic equipment according to the present embodiment.
(83) A digital camera **500** as an imaging apparatus of an example of the electronic equipment according to the present embodiment is, for example, a digital single lens reflex camera, a digital mirrorless camera, or the like. As illustrated in FIG. **6**, the digital camera **500** includes a camera body **200** and an interchangeable lens **300** that can be attached to and detached from the camera body **200**. In FIG. **6**, the interchangeable lens **300** is mounted on the camera body **200**. Hereinafter, a case where the interchangeable lens **300** is mounted on the camera body **200** to configure the digital camera **500** as an imaging apparatus will be described.
(84) The camera body **200** includes a housing **201**, an imaging unit **400**, and a circuit board **202**. The imaging unit **400** and the circuit board **202** are arranged inside the housing **201**. The camera body **200** may include a mirror, a shutter, and the like. The circuit board **202** is, for example, a printed wiring board and includes an image processing circuit. The camera body **200** includes a display device **203** fixed to the housing **201** so as to be exposed to the outside from the housing **201**. The display device **203** may be a liquid crystal display, for example.
(85) The interchangeable lens **300** includes a housing **301**, which is an interchangeable lens housing, and an imaging optical system **311**. The imaging optical system **311** includes a plurality of lenses. The housing **301** of the interchangeable lens **300** includes a lens-side mount having an opening. On the other hand, the housing **201** of the camera body **200** includes a camera-side mount having an opening. By fitting the lens-side mount and the camera-side mount together, the interchangeable lens **300** (housing **301**) is attached to the camera body **200** (housing **201**).
(86) The imaging optical system **311** is arranged inside the housing **301**, and forms an optical image on the imaging element **27** mounted on the electronic module **100** when the housing **301** is mounted on the housing **201**. The electronic module **100** is attached to the camera shake correction unit **410**, and forms the imaging unit **400**. The camera shake correction unit **410** functions as a driving unit that moves the printed wiring board **10** of the electronic module **100** inside the housing **201** to correct camera shake. The other of the flexible printed circuit boards **1** connected to the imaging unit **400** is connected to a connector **204** mounted on another circuit board **202**. The circuit board **202** is connected to the display device **203** through another flexible printed circuit board **205**.
(87) The imaging element **27** in the imaging unit **400** is an imaging element such as a CMOS image sensor, a CCD image sensor, or the like that photoelectrically converts an optical image provided by the imaging optical system **311**.
(88) According to the present embodiment, also in the digital camera **500**, it is possible to improve

the heat dissipation characteristic for dissipation of heat generated in the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10**, and suppress the influence of heat on the peripheral components **40**. Thus, the performance of the digital camera **500** can be improved.

(89) In the present embodiment, the interchangeable lens **300** is attached to the camera body **200**, but the present invention is not limited thereto. When the interchangeable lens **300** is only the camera body **200** which is not mounted, the camera body **200** is an imaging apparatus. In the present embodiment, the digital camera **500** is divided into the camera body **200** and the interchangeable lens **300**, but the present invention is not limited thereto. The digital camera **500** may be an integrated camera in which a lens is incorporated in the camera body **200**. Further, in the present embodiment, the digital camera **500** as an imaging apparatus of the electronic equipment has been described, but the present invention is not limited thereto. The electronic equipment may be any type of equipment that may include the electronic module **100**.

Example 1

(90) As an imaging unit of Example 1, the imaging unit **400**, in which the printed wiring board **10** has the configuration illustrated in FIGS. **4**A to **4**C in the configuration of the imaging unit **400** according to the fifth embodiment illustrated in FIGS. **5**A and **5**B, the printed wiring board **10**, was manufactured.

(91) In the imaging unit **400** of the first embodiment, the frame **25** was made of resin and has a thickness of 2 mm. As the imaging element **27**, a CMOS image sensor having a planar shape of the rectangle of 30 mm×20 mm was used. The cover glass **26** had a planar shape of the rectangle of 28 mm×38 mm.

(92) As the flexible printed circuit board **1**, the flexible base **2** and the coverlay **4** were made of polyimide, and the flexible wiring layer **3** and the first electrode **71** were made of Cu. The thickness of each of the flexible base **2** and the coverlay **4** was 12.5 μm, the thickness of the flexible wiring layer **3** was 12 μm, and the width of the flexible printed circuit board **1** was 17 mm.

(93) As the printed wiring board **10**, a printed wiring board in which the material of the printed wiring base **11** was a glass epoxy material and the materials of the conductive layers **12**, **13**, **14**, and **15** and the second electrode **72** were each Cu was used. The thickness of each of the conductive layers **12**, **13**, **14**, and **15** and the second electrode **72** was about 30 μm, the solder resist layer **17** was a film resist, and the thickness of the solder resist layer **17** was 20 μm. As an adhesive for fixing the printed wiring board **10** to the frame **401**, which was a metal frame, a UV curing resin was used. As the frame **401**, a frame having an outer shape of 50 mm×60 mm was used.

(94) The first electrodes **71** of the flexible printed circuit board **1** and the second electrodes **72** of the printed wiring board **10** were connected by the solder **5**. The pitch of the second electrodes **72** was 0.2 mm, the electrode width was 0.1 mm, and the number of electrodes were **80**. The opening of the solder resist layer **17** where the second electrode **72** was exposed was set to have a size of 1.0 mm×20 mm. On the other hand, the pitch of the first electrodes **71** was 0.2 mm, the electrode width was 0.1 mm, and the number of electrodes were **80**. The pitch of the electrodes, the electrode width, and the number of the electrodes were appropriately set in accordance with the specifications of the imaging unit **400**.

(95) The short side of the reference region **50** between the flexible printed circuit board **1** and the printed wiring board **10** was 1 mm, and the long side thereof was 16 mm. The first region **51** was a region of a rectangle having a short side length L**1** of 3 mm and a long side length W**1** of 20 mm. The second region **52** was a region excluding the first region **51** from a region of a rectangle having a short side of 6 mm and a long side of 26 mm.

(96) The circular openings **20** of the second region **52** were arranged periodically in a staggered manner. Specifically, a line of similar opening groups adjacent in the Y direction to a line of opening groups including a plurality of openings **20** arranged in the X direction was shifted by +0.3 mm in the X direction, and this was periodically repeated. The planar shape of the opening **20** was

circular with a diameter of 0.3 mm. The pitch of the openings **20** in the X direction was 0.6 mm, and the pitch of the openings **20** in the Y direction was 0.3 mm.

(97) The elliptical openings **21** of the second region **52** were periodically arranged in a staggered manner. Specifically, a line of similar opening groups adjacent in the +Y direction to a line of opening groups including a plurality of openings **21** arranged in the X direction was shifted by +1 mm in the X direction, and this was periodically repeated. The planar shape of the opening **21** was an elliptical shape having a short axis of 0.5 mm and a long axis of 1.0 mm. The pitch of the openings **21** in the X direction was 2.0 mm, and the pitch of the openings **21** in the Y direction was 1.0 mm.

(98) The rectangular opening **90** formed in the first region **51** had a short side length L**5** of 0.3 mm and a long side length L**4** of 1.5 mm. One of the short sides of the opening **90** was in contact with the boundary line of the reference region **50**. The distance between the openings **90** adjacent to each other in the X direction was 0.3 mm. Three openings **90** were arranged.

(99) As the peripheral component **40**, an IC of 4 mm square, which is an active component, was used. The solder **5** made of Sn-3.0Ag-0.5Cu was used. As the camera shake correction unit **410**, a unit having an L shape in which a rectangle of 70 mm×55 mm was cut out from the rectangle of 85 mm×70 mm was used.

(100) The imaging apparatus equipped with the completed imaging unit **400** of Example 1 was able to sufficiently ensure the optical performance of the imaging element **27** which was a built-in CMOS image sensor.

Example 2

(101) As an imaging unit of Example 2, the imaging unit **400**, in which the printed wiring board **10** has the configuration illustrated in FIG. **3** in the configuration of the imaging unit **400** according to the fifth embodiment illustrated in FIGS. **5**A and **5**B, the printed wiring board **10**, was manufactured.

(102) In the imaging unit **400** of Example 2, the heat dissipation part **31** was arranged on the bonded portion between the flexible printed circuit board **1** and the printed wiring board **10** by the solder **5**. The protruding portions **32**, which were protruding projections, were formed on parts of the flexible printed circuit board **1** in the width direction, and a resin was applied as the reinforcing part **30** from above to fix each of the protruding portions **32** to the printed wiring board **10**.

(103) The same flexible printed circuit board as in Example 1 was used as the flexible printed circuit board **1**. As the printed wiring board **10**, the same one as in Example 1 was used except that the rectangular opening **90** and the elliptical opening **21** of the first region **51** were not formed.

(104) The graphite sheet was used as the heat dissipation part **31**. The shape of the heat dissipation part **31** was a rectangular parallelepiped having a long side of 20 mm, a short side of 10 mm, and a height of 2.5 mm. Each of the protruding portion **32** of the flexible printed circuit board **1** was formed at a position of 0.5 mm in a direction opposite to the tip of the first electrode **71** of the flexible printed circuit board **1** from the end of the first electrode **71** of the flexible printed circuit board **1** on the coverlay **4** side. That is, each of the protruding portion **32** was formed at a position of 1.5 mm in the length direction of the flexible printed circuit board **1** from the tip of the first electrode **71**. The reinforcing part **30** made of a semi-elliptic spherical ultraviolet curing resin having a diameter of about 3 mm and a height of about 1 mm was formed on each of the protruding portion **32**. The heat dissipation part **31** was connected to a sheet metal in the housing. The sheet metal also functioned as a heat dissipation part.

(105) The imaging apparatus equipped with the completed imaging unit **400** of Example 2 was able to sufficiently ensure the optical performance of the imaging element **27** which was a built-in CMOS image sensor. Further, in Example 2, since the heat dissipation part **31** was added, the heat generated in the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10** was able to be dissipated, and the influence on the peripheral component **40** was able to be further reduced. Further, in Example 2, the protruding portions **32** were formed on the

flexible printed circuit board **1**, and the protruding portions **32** were reinforced by the reinforcing parts **30** made of an ultraviolet curing resin. Therefore, in Example 2, the load on the connected portion **6** between the flexible printed circuit board **1** and the printed wiring board **10** was able to be reduced simultaneously with countermeasures against heat.
(106) According to the present invention, in the electronic module in which the flexible printed wiring board is connected to the printed wiring board by soldering, it is possible to provide a technique capable of suppressing the influence of heat on peripheral component mounted on the printed wiring board to be small and improving the heat dissipation performance.
(107) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(108) This application claims the benefit of Japanese Patent Application No. 2023-011144, filed Jan. 27, 2023, which is hereby incorporated by reference herein in its entirety.

Claims

1. An electronic module comprising: a flexible printed circuit board having a first electrode group; a printed wiring board having a second electrode group; and an electronic component mounted on the printed wiring board, wherein the printed wiring board includes a first conductive layer formed to include the second electrode group and a second conductive layer formed under the first conductive layer, wherein the first electrode group and the second electrode group are bonded by solder, wherein a plane including the second conductive layer includes a first region, a second region in contact with the first region via a first boundary, and a third region in contact with the second region via a second boundary in a plan view with respect to the plane, wherein the first region includes a reference region having a contour of a rectangle, and the reference region is set such that four sides of the rectangle are circumscribed to a bonded portion where the first electrode group and the second electrode group are bonded to each other in the plan view, wherein a first boundary portion, which is at least a part of the first boundary, is positioned within a range of 0.5 times or more and 5 times or less a length of a short side of the four sides in one direction from one side of the four sides of the reference region toward the outside of the reference region, wherein a second boundary portion, which is at least a part of the second boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in the one direction from the first boundary portion, and wherein a density of the second conductive layer in the first region is higher than a density of the second conductive layer in the second region, and the electronic component is arranged on the third region.
2. The electronic module according to claim 1, wherein a density of the second conductive layer in a region excluding the reference region from the first region in the plan view is higher than a density of the second conductive layer in the reference region.
3. The electronic module according to claim 1, wherein a plurality of openings are formed in the second conductive layer in the second region.
4. The electronic module according to claim 1, wherein the plurality of openings includes a first opening and a second opening having an opening area smaller than that of the first opening.
5. The electronic module according to claim 4, wherein the second conductive layer is continuous from the first opening to the first region, and wherein the second conductive layer is continuous from the second opening to the first region.
6. The electronic module according to claim 4, wherein the first opening is closer to a center of a long side of the reference region than an edge of the long side.
7. The electronic module according to claim 1 comprising: an electronic component arranged on the second region, wherein the electronic component arranged on the second region has higher heat

resistance than the electronic component arranged on the third region.

8. The electronic module according to claim 1, wherein the first boundary portion is positioned within a range of 1 times or more and 2 times or less the length of the short side in the one direction from the one side.

9. The electronic module according to claim 1, wherein the second boundary portion is positioned within a range of 1.5 times or more and 2.5 times or less the length of the short side in the one direction from the first boundary portion.

10. The electronic module according to claim 1, wherein the density of the second conductive layer in the first region is 50% or more in area ratio.

11. The electronic module according to claim 1, wherein a third opening is formed in the second conductive layer in the first region.

12. The electronic module according to claim 11, wherein the third opening is formed so as to overlap with a second electrode included in the second electrode group in the plan view.

13. The electronic module according to claim 12, wherein the third opening has a plane shape having a short direction and a longitudinal direction, wherein at least one of a length of the short direction of the third opening being within range of 1 times or more and 3 times or less the length of a short direction of the first electrode included in the first electrode group, and a length of the longitudinal direction of the third opening being within range of 1 times or more and 2 times or less the length of a longitudinal direction of the first electrode, is satisfied.

14. The electronic module according to claim 1, wherein the one side is a long side of the four sides.

15. The electronic module according to claim 1, wherein the second region and the flexible printed circuit board overlap each other in the plan view.

16. The electronic module according to claim 1, wherein the one side is a first side and the one direction is a first direction, wherein a third boundary portion, which is at least a part of the first boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in a second direction from at least a second side of the four sides of the reference region to the outside of the reference region, and wherein a fourth boundary portion, which is at least a part of the second boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in the second direction from the third boundary portion.

17. The electronic module according to claim 16, wherein a fifth boundary portion, which is at least a part of the first boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in a third direction from at least a third side of the four sides of the reference region to the outside of the reference region, and wherein a sixth boundary portion, which is at least a part of the second boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in the third direction from the fifth boundary portion.

18. The electronic module according to claim 17, wherein a seventh boundary portion, which is at least a part of the first boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in a fourth direction from at least a fourth side of the four sides of the reference region to the outside of the reference region, and wherein an eighth boundary portion, which is at least a part of the second boundary, is positioned within a range of 0.5 times or more and 5 times or less the length of the short side in the fourth direction from the seventh boundary portion.

19. The electronic module according to claim 1 comprising a heat dissipation part that covers the bonded portion, wherein the flexible printed circuit board is provided between the heat dissipation part and the printed wiring board so that heat from the bonding portion is transferred to the heat dissipation part via the flexible printed circuit board.

20. The electronic module according to claim 1 comprising an imaging element mounted on the printed wiring board.

21. Equipment comprising: a housing; the electronic module according to claim 1 arranged inside the housing; and a driving unit configured to move the printed wiring board inside the housing.